



US Health Insurance data

Camille Rouan & Sacha Sagratella

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1 Introduction

The goal of this project is to determine what is the extreme value behavior of a medical claim amount. We use a dataset containing the value of US medical insurance claims in 1991. The dataset of 75,788 observations only contains claims exceeding \$25,000, meaning that the bulk of small claims is unobserved. Consequently, the analysis focuses on tail modeling rather than full distribution modeling.

Medical insurance claims are well-known to show heavy-tailed behavior: while a large majority of claims remain moderate, a small number of severe cases can drive enormous aggregate losses. Therefore, understanding the tail of the claim distribution is very important for insurers. Extreme Value Theory (EVT) provides a rigorous statistical framework for this purpose, enabling principled inference on rare, high-severity events that lie beyond the bulk of observed data.

This report is structured as follows. Section 2 reports the construction of the extreme value analysis using techniques from the course (Hill estimator, extrapolation ect.), Section 3 checks for robustness in results, Section 4 studies the average magnitude of extreme insurance claims and finally Section 5 presents a general model describing both the bulk and the tail of the data, before concluding.

2 Extreme Value Theory

Exploratory Analysis

A first brief statistical analysis suggests (Table 1) the presence of extreme values. While the median is equal to 40223 and the third quartile to 61322, the mean seems to be pulled up (58413) by some extreme positive values. Indeed, the empirical maximum is equal to 4,518,420. This reflects in the kurtosis (487) suggesting the presence of very large observations and thus a heavy-tail.

Min	1st qu.	Median	Mean	3rd qu.	Max	Skew.	Kurt.
25000	30543	40223	58413	61322	4518420	13.209	487.589

Table 1: Descriptive Statistics

These first observations are consistent with the histograms in Figure 1. We observe some very extreme observations on the right far tail highlighted by the red rug on the histogram in level. Because of these extremes, we also consider the logarithm of the data. The log-scale histogram becomes more parsimonious. In addition, we use boxplot (in level and log), reported in Figure 6, to have another point of view of the extremes.

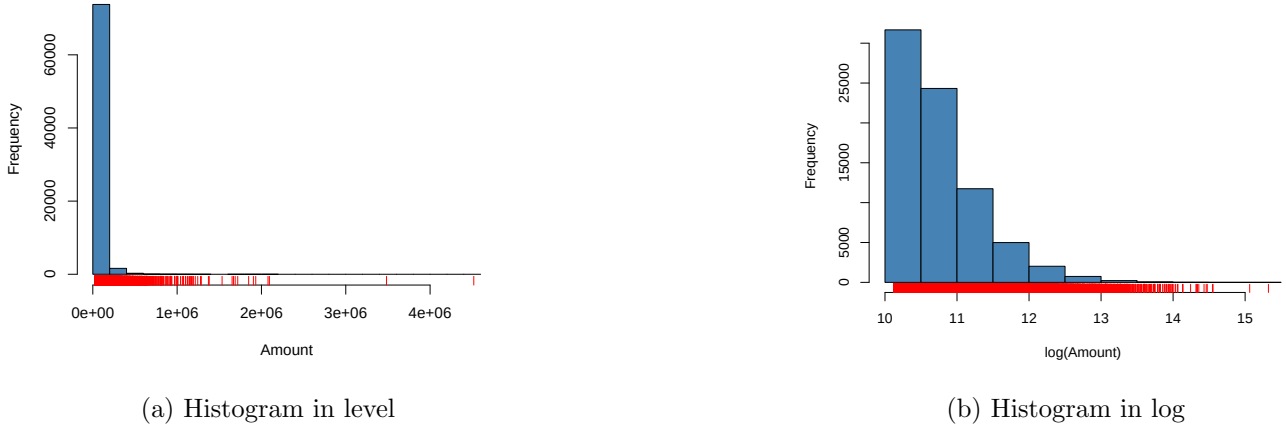


Figure 1: Histogram of the Amount of claims: level vs log

The boxplot on level is dominated by extreme observations, making the central distribution very crushed. After applying the log, the spread of the data becomes clearer as it was the case for the histogram.

Even if we suspect a heavy-tail distribution, the first step is to fit some distributions in order to check if they are convenient for the data. We tried different ones, but in order to avoid redundancy, we will only report the fit of the Log-normal and Weibull distributions (Figure 7 and Figure 8).

In both cases, we observe similar scenarios. For the bulk of the data, the Q-Q plots seem to provide a reasonable fit. However, it fails to capture the extreme upper tail. Indeed, we observe the empirical quantiles deviating from the empirical ones when approaching the tail. Both P-P plots suggest a poor overall fit.

Hill Estimator

Usual distributions do not fit the data as they fail to capture extreme observations, i.e. tail behavior. Moreover, the data seems to be heavy-tailed based on the descriptive statistics and the first visual inspection done.

In that context, we decide to use Extreme Value Theory (EVT). EVT uses extrapolation of intermediate quantile in order to determine extreme quantile.

$$q(\beta) = \left(\frac{1-\beta}{1-\alpha} \right)^{-\gamma} \cdot q(\alpha),$$

with $q(\beta)$ the extreme quantile, $q(\alpha)$ the intermediate one and $\left(\frac{1-\beta}{1-\alpha} \right)^{-\gamma}$ the extrapolation factor. $\hat{q}(\alpha) = \hat{q}_n(1-k/n) = X_{n-k,n}$ is the empirical quantile. The crucial step is then to set the level of an intermediate quantile and determine the threshold exceeded by k observations. To do so, we use the Hill estimator to determine γ :

$$\hat{\gamma}_k = \frac{1}{k} \sum_{i=1}^k \log \frac{X_{n-i+1,n}}{X_{n-k,n}}$$

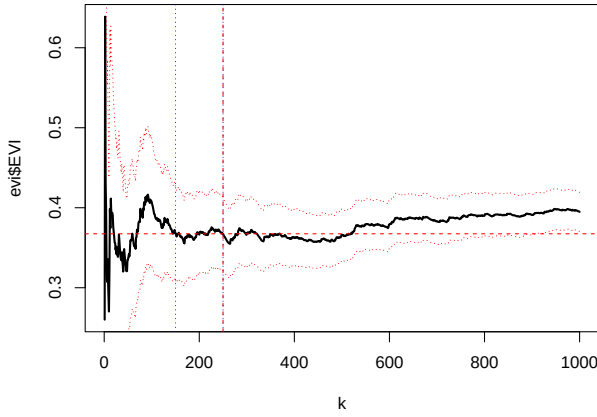
Figure 2 represents the Hill estimator according to the level of exceedances k , we plot the confidence intervals from

$$CI = \hat{\gamma} \pm 1.96 \times \frac{\hat{\gamma}}{\sqrt{k}}.$$

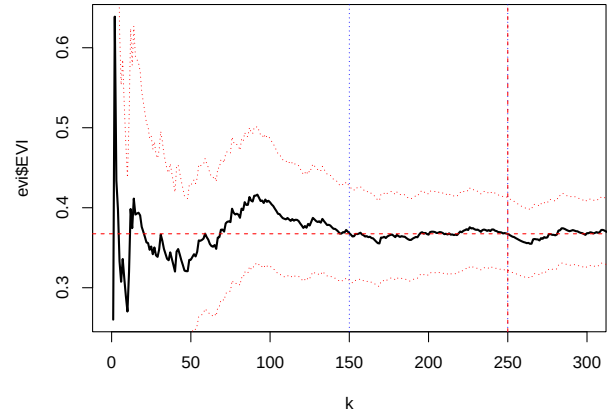
We select k as the last value of the first stable zone. Here, we identify this first zone between $k = 150$ and 250 , leading to set k to 250 . Thus, the Hill estimator is $\hat{\gamma}_{250} = 0.367$.

Moreover, there are three domains of attraction (i.e. classes of distributions) following the sign of γ :

- The Fréchet domain of attraction, or set of heavy-tailed distributions, for which $\gamma > 0$,
- The Weibull domain of attraction, or set of short-tailed distributions, for which $\gamma < 0$,
- The Gumbel domain of attraction, or set of light-tailed distributions, for which $\gamma = 0$.



(a) Hill plot full



(b) Hill plot focus on the stable region

Figure 2: Hill estimator

We observe $\hat{\gamma}_k > 0$, which confirms our first assumption that the data is heavy-tailed and then belongs to the Fréchet domain of attraction. Notice that $\gamma < 0.5$ indicating that moments are finite.

If k is chosen well and the Pareto approximation is reasonable, then

$$E_i^* = i \times \log \frac{X_{n-i+1,n}}{X_{n-k,n}}$$

with $1 \leq i \leq k$, should behave approximately like i.i.d. exponentials with mean γ . Therefore, to check if we selected the right k we use an exponential Q-Q plot: theoretical quantiles $\hat{\gamma}_k \log \frac{k+1}{k+1-i}$ at levels $1/(k+1), \dots, k/(k+1)$ of the exponential distribution with mean γ versus the empirical quantiles $E_{i,k}^*$ of the observations E_i^* . We can observe in Figure 3 that few extreme points are not aligned with the diagonal, but most of them are. We can deduce that our choice of k looks appropriate. In addition, we plot a Pareto Q-Q plot in Figure 9 and observe similar results.

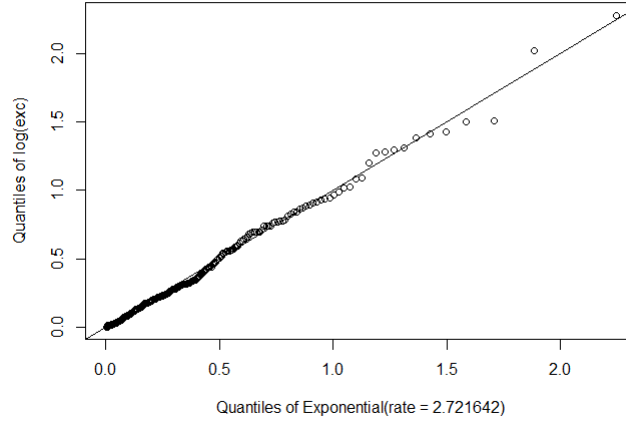


Figure 3: Exponential Q-Q plot

Extrapolation

The objective of extrapolation is to estimate very high quantiles corresponding to extremely rare medical claims in US insurance. EVT provides models for exceedances, enabling principled extrapolation beyond the observed range of data. We proceed to the extrapolation by applying the formula defined before. In our case, $\alpha = 1 - \frac{k}{n} = 1 - \frac{250}{75788} = 0.9967$. Therefore, we take different values for $\beta = 0.99$; 0.995 ; 0.999 and the very extreme level $1 - \frac{1}{n}$. In addition, we compute the return periods for each β by using $\frac{1}{1-\beta}$. For instance, the 99.9% quantile corresponds to a claim expected to be exceeded once every 1,000 claims on average. The highest level ($\beta = 0.9999868$) represents an event occurring approximately once over the whole sample size (once over 75,788!). The results are presented in [Table 2](#).

β	Return Period	Empirical Quantile	EVT Extrapolated Quantile	Difference
0.99	100	305 970	313 648	7 678
0.995	200	406 228	404 622	-1 606
0.999	1 000	721 135	730 917	9 782
0.9999868	75 788	3 483 562	3 584 933	101 371

Table 2: Extreme quantile extrapolation and associated return periods

The EVT extrapolated quantiles are very close to the empirical ones for high but observable probability levels ($\beta = 0.99, 0.995, 0.999$). This validates the adequacy of the tail model. For the extremely high level ($\beta = 0.9999868$), where empirical estimation becomes unreliable due to the lack of observations, EVT provides a slightly stable extrapolation beyond the observed range with a difference of $\sim \$100,000$ on a claim of $\sim \$3.5M$ which occurs, in average, once every 75,788.

However, the extrapolated quantile of the very extreme level of $\beta = 1 - \frac{1}{n}$ is still lower than the observed maximum in the data which is \$4,518,420. This is not contradictory. Indeed, the return level $q(1 - 1/n)$ represents the theoretical level expected to be exceeded once every n observations on average, whereas the sample maximum is a highly random variable. For heavy-tailed distributions, large fluctuations of

the maximum are expected, and some realizations could exceed the theoretical return level. [Figure 10](#) illustrates that.

3 Robustness Checks

As shown in [Figure 10](#), the observed maximum lies on the upper bound of the confidence interval, therefore we should check the robustness of our estimates. First, we will do the same analysis without the observed maximum, and then, we will remove the 5 highest values. This will allow us to check if our results are sensitive to deleting part of the data.

Removing top observations

We use three different datasets:

- Full original dataset
- original dataset without the sample maximum
- original dataset without the top 5 observations

On these three databases we did the same procedure as before to obtain the Hill plots for each of the them, as illustrated in [Figure 11](#). We can observe that removing observation(s) makes the value of the Hill estimator lower. Then, at $k = 250$ (as selected in Section 1) the value of $\hat{\gamma}$ is 0.3674 ; 0.3585 and 0.3347 with the full data, without the max and without top 5 values, respectively ([Table 3](#)). The Hill estimator remains slightly stable while the extrapolated quantiles are close when removing the maximum (3.58M vs 3.41M), but they are much further when removing the top 5 (3.58M vs 2.98M). This suggests that the top 5 maxima have an impact on the magnitude of the extreme losses but not on tail heaviness. Moreover, the confidence intervals overlap each others, meaning that our model is robust against the deletion of maxima.

Dataset	n	$\hat{\gamma}$	Extrapolated Quantile	CI Lower	CI Upper
Full data	75 788	0.3674	3 584 933	2 683 399	4 486 467
Without maximum	75 787	0.3585	3 412 322	2 575 016	4 249 629
Without top 5 values	75 783	0.3347	2 986 284	2 302 255	3 670 313

Table 3: Sensitivity analysis of the tail index and extrapolated extreme quantile

Variation of k

As another robustness check, we select three different values of k : 200 ; 250 and 300. We obtain the results shown in [Table 4](#). The estimated EVI and extrapolated extreme quantiles remain very stable when varying the number of exceedances k . This stability confirms the presence of a reliable stable zone in the Hill plot and supports the validity of the chosen threshold. Hence, the extreme risk estimates do not critically depend on the specific selection of k .

k	$\hat{\gamma}$	Threshold	Extrapolated Quantile	CI Lower	CI Upper
200	0.3663	512 458	3 569 628	2 609 385	4 529 871
250	0.3674	471 430	3 584 933	2 683 399	4 486 467
300	0.3683	440 788	3 601 105	2 745 188	4 457 023

Table 4: Sensitivity analysis with respect to the number of exceedances k

4 Extreme Insurance Claims: Average Magnitude

The goal of this section is to estimate the Mean Excess Function (MEF):

$$e(u) = \mathbb{E}[X \mid X > u]$$

with u a high threshold, indicating a large claim in our case. This allows us to know if we have a claim above u , then on average how large it is. Under the Generalized Pareto approximation, the MEF admits the linear form $e(u) = \frac{\sigma + \gamma u}{1 - \gamma}$.

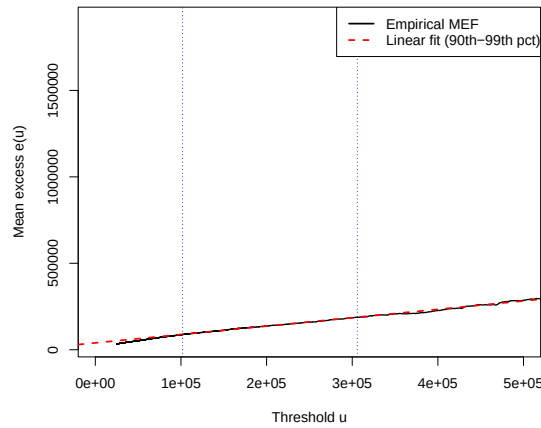


Figure 4: Mean Excess Function

We can observe in [Figure 4](#) an increasing and approximately linear MEF for high thresholds u as expected for a heavy tailed function. Larger claims tend to be followed by even larger expected excess losses, therefore, once a claim becomes severe, its expected additional cost increases highlighting the presence of substantial tail risk. The linear behavior observed in the plot is consistent with the positive Hill estimator obtained previously. Indeed, both diagnostics independently indicate heavy-tail behavior and mutually reinforce the validity of the EVT framework.

Then, we compare the empirical MEF with the theoretical Pareto approximation for several thresholds. We observe a strong agreement between empirical and theoretical values, especially for high thresholds, indeed, the empirical MEF converges to the Pareto estimation as u increases. This supports the validity of the Pareto tail approximation and confirms that the selected EVT threshold lies in the region where Extreme Value Theory assumptions are appropriate.

Threshold Level	Threshold u	n_u	Empirical MEF	Pareto Approximation
$q(0.95)$	147 563	3 790	258 455	233 273
$q(0.99)$	305 970	758	494 014	483 690
$q(0.999)$	721 135	76	1 151 879	1 139 999
$q(1 - k/n)$	471 430	250	741 694	745 256

Table 5: Empirical and Pareto mean excess comparison at different thresholds

Table 5 reports the results of the MEF estimation for different thresholds. For the EVT threshold $1 - \frac{k}{n}$ we use again $k = 250$ and n_u reports the number of observations above the threshold u . Then we compare the estimation of the empirical MEF versus the Pareto approximation.

Looking at the results, the higher u the more similar the results are between empirical and the EVT. Indeed, $q(0.95)$ is not yet very extreme as it is still in the "bulk" of the data. As EVT does not work well non extreme values (i.e. the bulk), the observed difference is not significant. For the higher thresholds (i.e. at 0.99, 0.999 and $1-k/n$), the results are very similar between the two estimations. For instance, the EVT threshold is empirically equal to $\sim 742k$ versus $\sim 745k$ for the Pareto approximation.

Focusing on the results of the EVT threshold we compute the confidence intervals for the conditional expectation of the EVT threshold. This gives

$$\widehat{\mathbb{E}}[X \mid X > u_{\text{EVT}}] = 745\,256, \quad 95\% \text{ CI} = [691\,597; 798\,915].$$

These results indicate that, once a medical claim exceeds the EVT threshold of \$471k, the expected claim size is approximately \$745k on average. The CI is relatively short suggesting stable estimation and confirms the adequacy of the GPD approximation for modeling large insurance losses. Moreover, it remains well above the threshold level of \$471k, confirming the significativity of the EVT model to assess extreme medical claims.

5 General model: Central and Extreme Behavior

In this final section, we want to find a statistical model that describes the central and extreme value behavior of the claim amount distribution at the same time.

The insurance data has, in general, a lot of moderate values and few ones very extremes leading to fat tails. As shown earlier, our US health insurance data is heavy-tailed. Therefore, the bulk and the tail of the data have very different behaviors, and using GPD or Pareto models will not work well for the central behavior of the distribution. The idea is then to use a *spliced model* which combines a model for the bulk and another one for the tail.

Let t denote a high threshold selected from the EVT analysis, defined as $t = q(1 - \frac{k}{n}) = q(0.9967013) = \$471,430$, where k represents the number of exceedances used in the Hill estimation and n the sample

size. This threshold separates the distribution into two distinct regions:

$$X = \begin{cases} X_{\text{bulk}}, & X \leq t, \\ X_{\text{tail}}, & X > t. \end{cases}$$

We therefore construct the *spliced model* using the threshold t . For observations below t , the bulk distribution is modeled using a truncated parametric distribution defined on the interval $[x_{\min}, t]$, where $x_{\min}$ denotes the minimum observed claim (\$25000). We naturally used a truncated Lognormal distribution for the bulk of the data. For exceedances above t , we apply the Peaks-Over-Threshold framework and model excess losses using a Generalized Pareto Distribution (GPD), estimated using Maximum Likelihood. The bulk proportion is $\phi = P(X \leq t) = 0.9967$

Results

The main results are presented in [Table 6](#) as well as diagnostic plots (density and QQ-plot) in [Figure 5](#). We observe similar results for the tail, compared when we did the Weissman estimation: $\hat{\gamma}_{GPD} = 0.376$ vs $\hat{\gamma}_{Weiss} = 0.367$. Again, this confirms the heaviness of the tail, as $\hat{\gamma}$ is positive. Notice that the scale parameter is very high ($\hat{\sigma}_{GPD} = 171,878$), which means that the distribution is concentrated at the far right: indeed, extreme claims (located in the right tail) are very severe. On the contrary, the average claim of the bulk of the data is equal to $\exp(\mu + \frac{\sigma^2}{2}) \approx \4432.85 . Indeed, extreme observations had a big impact on the overall distribution: taking into account the tails, the average claim was equal to \$58413, which is much higher.

We observe that the GPD and the Weissman extreme quantile estimates are quite close (3 658 974 vs 3 584 933), which would suggest that the tail behavior was well captured, and that the choice of the threshold is appropriate. Moreover, as both models lead to close results, we could deduce that the tail approximately follows a Pareto.

Looking at [Figure 5](#), we observe that the spliced model is a reasonable fit. The density plot shows a good fit relative to the histogram, even if it does not capture well near the truncation zone. Overall, it seems to be a good fit. Most of the observations on the Q-Q plot are very close to the $y = x$ line. More precisely, observations are very well aligned on the axis for the bulk of the data. Some observations (extreme ones) drift apart from the $y = x$ axis, however remain close. Only very few observations are very far from it.

Overall, the spliced model is not a perfect fit but seems to still be a reasonable one. : the bulk component suffers from identifiability issues under truncated MLE, while the tail component performs comparably to the Weissman estimator. A more robust alternative for the bulk could improve the overall fit. Nevertheless, based on this observation we also fitted a Gamma distribution and compared models using Lognormal or Gamma distributions via AIC. Despite its numerical instability, the truncated lognormal yielded a lower AIC than the Gamma (1,701,040 vs 1,733,382), and was therefore retained as the bulk component of the spliced model. We can observe the comparison between Lognormal and Gamma distributions on the bulk of the data in [Figure 12](#) and [Figure 13](#).

Component	Parameter / Quantity	Estimate
<i>Bulk model (Truncated Lognormal)</i>		
Meanlog parameter	$\hat{\mu}$	7.069
Sdlog parameter	$\hat{\sigma}$	1.6296
Log-likelihood	ℓ_{bulk}	-850 517.8
AIC	AIC_{LN}	1 701 040
<i>Tail model (GPD – MLE)</i>		
Scale parameter	$\hat{\sigma}_{\text{GPD}}$	171 848
Shape parameter	$\hat{\gamma}$	0.376
Hill estimate (comparison)	$\hat{\gamma}_{\text{Hill}}$	0.3674
<i>Extreme quantile</i>		
Level	$\beta_n = 1 - \frac{1}{n}$	0.9999868
Spliced estimate	$\hat{q}_{\text{spliced}}(\beta_n)$	3 658 974
Weissman estimate	$\hat{q}_{\text{EVT}}(\beta_n)$	3 584 933
Observed maximum	$\max X$	4 518 420

Table 6: Summary of the spliced Lognormal–GPD model describing central and extreme claim behaviour

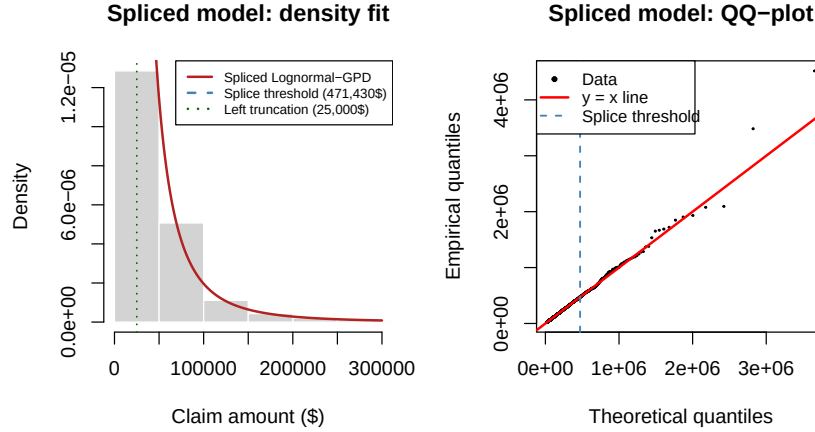


Figure 5: Diagnostics plots of the spliced model

6 Conclusion

This project investigated the extreme value behavior of US medical insurance claim amounts. The exploratory analysis suggests evidence of heavy-tailed distribution of the data behavior: a mean significantly above the median, very high kurtosis and a sample maximum exceeding \$4.5 million (while the median was at \$40,223). Standard parametric distributions such as the Log-normal and Weibull failed to capture well the far right tail of the data, motivating the use of Extreme Value Theory.

Applying the famous Hill estimator, we estimated the Extreme Value Index at $\hat{\gamma} = 0.367$, confirming that the claim distribution belongs to the Fréchet domain of attraction ($\hat{\gamma} > 0$). Using the Weissman estimator for extrapolated quantiles we found that they closely match empirical quantiles at observable levels, and provided stable estimates beyond the observed range.

Robustness checks confirmed the stability of the results. In fact, removing the top 1 or top 5 observa-

tions left the tail index largely unchanged, and varying the threshold k between 200 and 300 (versus 250 at the beginning) produced nearly identical estimates. This stability supports the reliability of our tail model and the use of EVT.

The Mean Excess Function analysis further corroborated the heavy-tail finding: the MEF is approximately linear and increasing, consistent with a Pareto tail. Once a claim exceeds the EVT threshold of \$471,430, the expected claim size rises to approximately \$745,000. This has concrete implications for insurers: large claims do not just exceed a threshold by a little, they tend to be significantly larger than it (almost twice as much!), reflecting a significant risk for insurance.

Finally, a spliced Lognormal-GPD model was constructed to jointly describe the central and extreme behavior of the distribution. The truncated Log-normal captures the bulk of claims (below \$471k), while the GPD models the tail exceedances. This model provides coherent but not perfect results, for instance the GPD shape parameter ($\hat{\gamma} = 0.376$) is consistent with the Hill estimate, and the extreme quantile produced by the spliced model (\$3.66M) closely agrees with the Weissman estimate (\$3.58M), confirming the coherence of the two approaches.

To wrap up, US medical insurance claim amounts exhibit indeed heavy-tailed behavior. EVT methods provide reliable tools for tail estimation and extreme risk quantification in this context.

The results highlight that extreme medical claims represent a substantial and predictable source of tail risk, with important implications for insurance and reinsurance.

AI acknowledgment

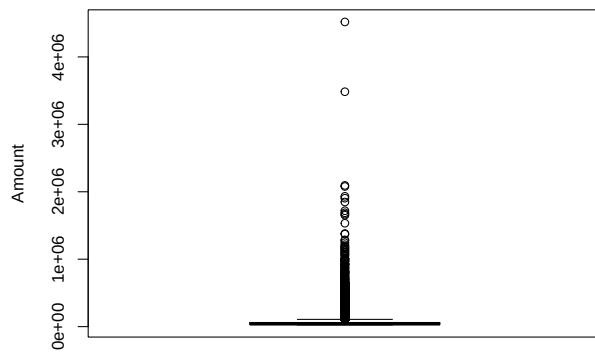
AI tools have been used for this report in some ways:

- help with code (errors, checking etc),
- help with LaTeX formatting (tables, equations, etc.),

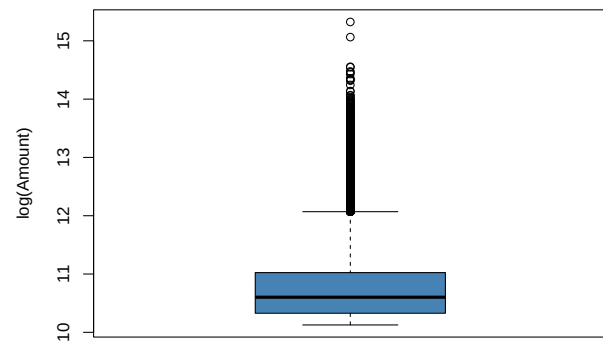
References

- [1] Abdelaati Daouia, *Extreme Risk Analysis* lecture notes (2026).
- [2] Abdelaati Daouia, *Extreme Risk Analysis* Practicals (2026).

Appendix



(a) Boxplot in level



(b) Boxplot in log

Figure 6: Boxplot: level vs log

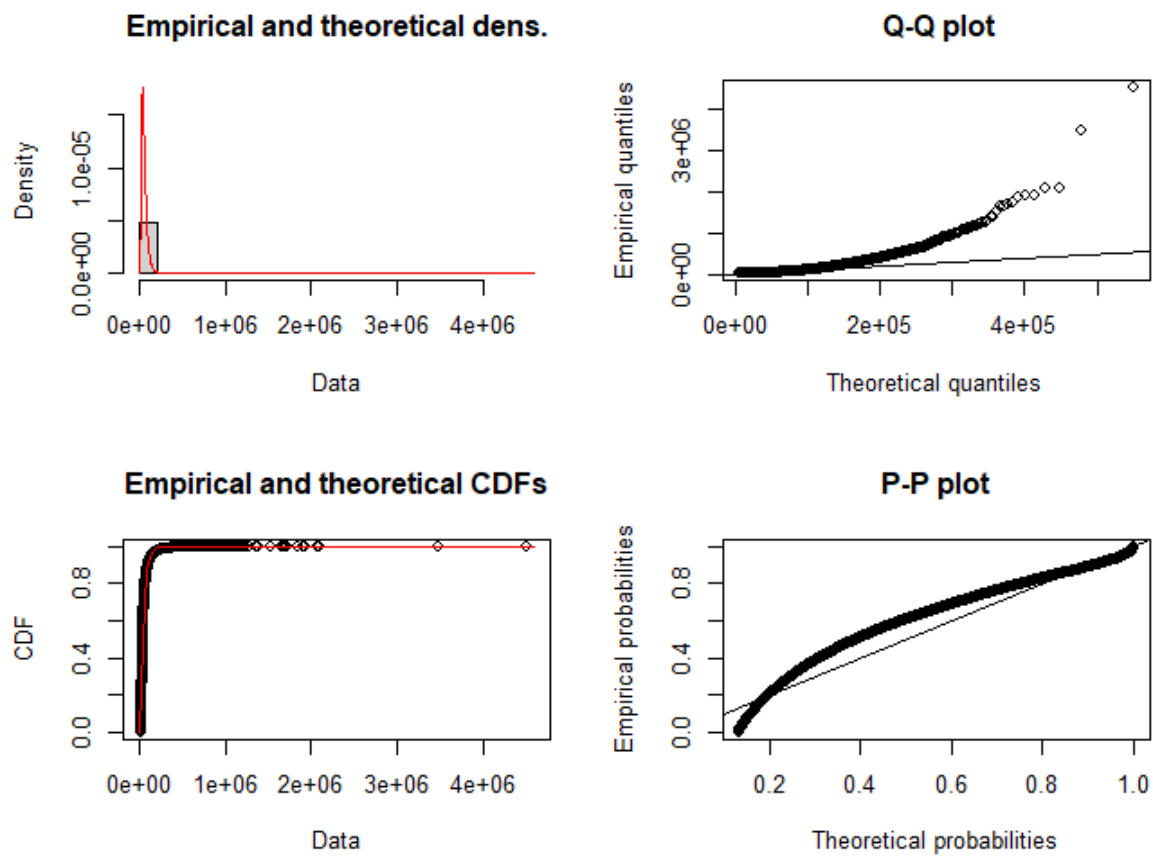


Figure 7: Lognormal fit

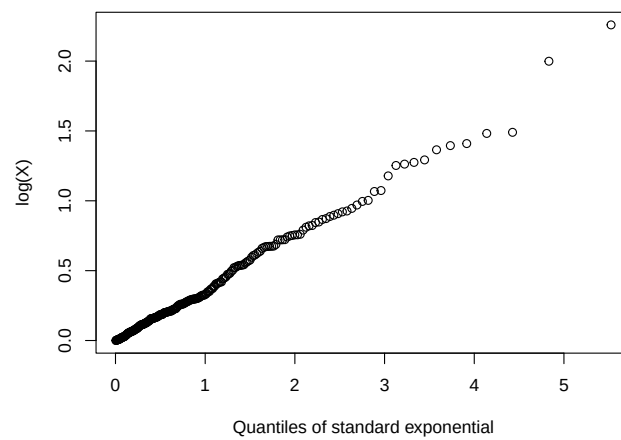


Figure 9: Pareto Q-Q plot

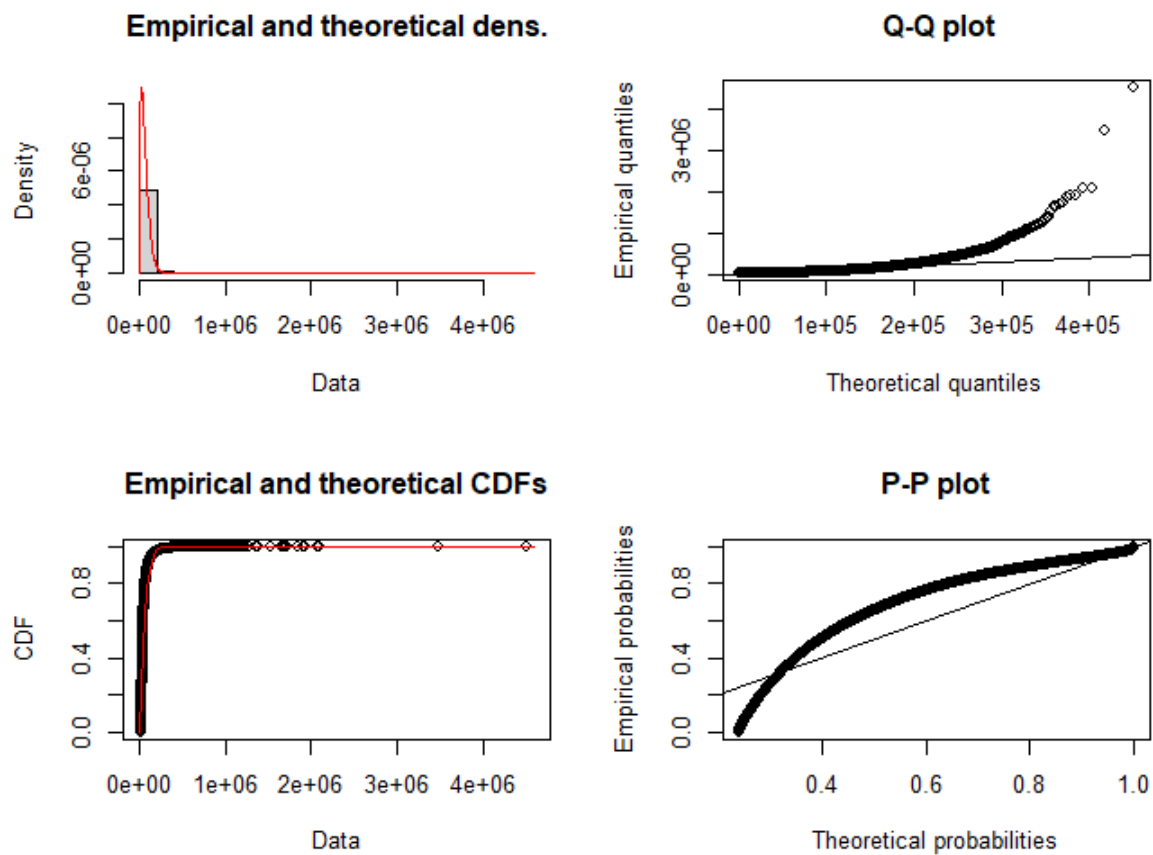


Figure 8: Weibull fit

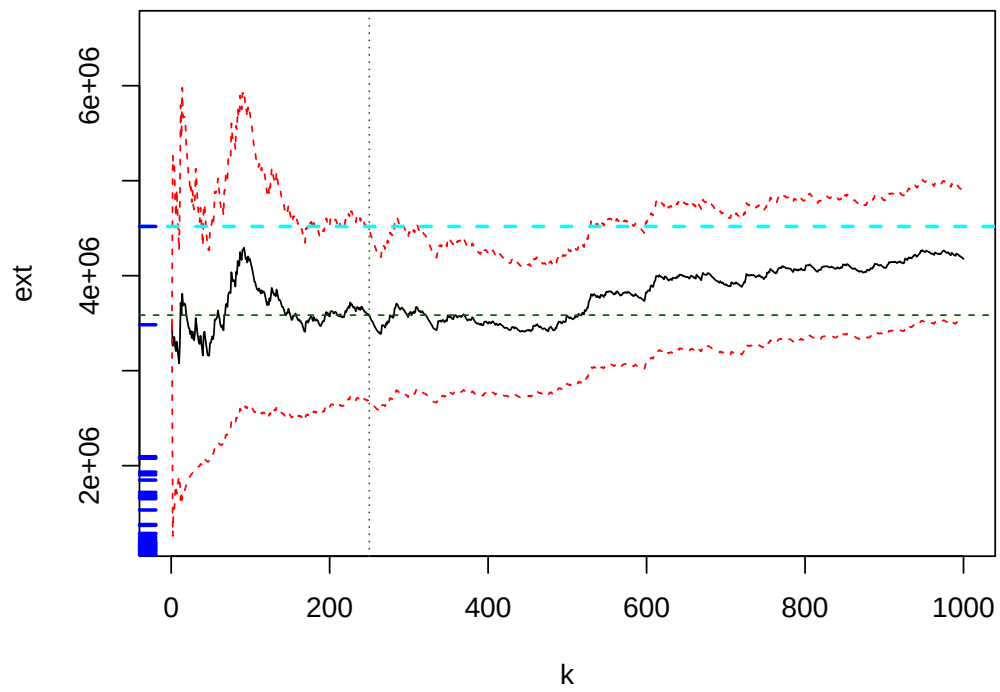


Figure 10: Observed maximum and Extrapolated extreme quantile

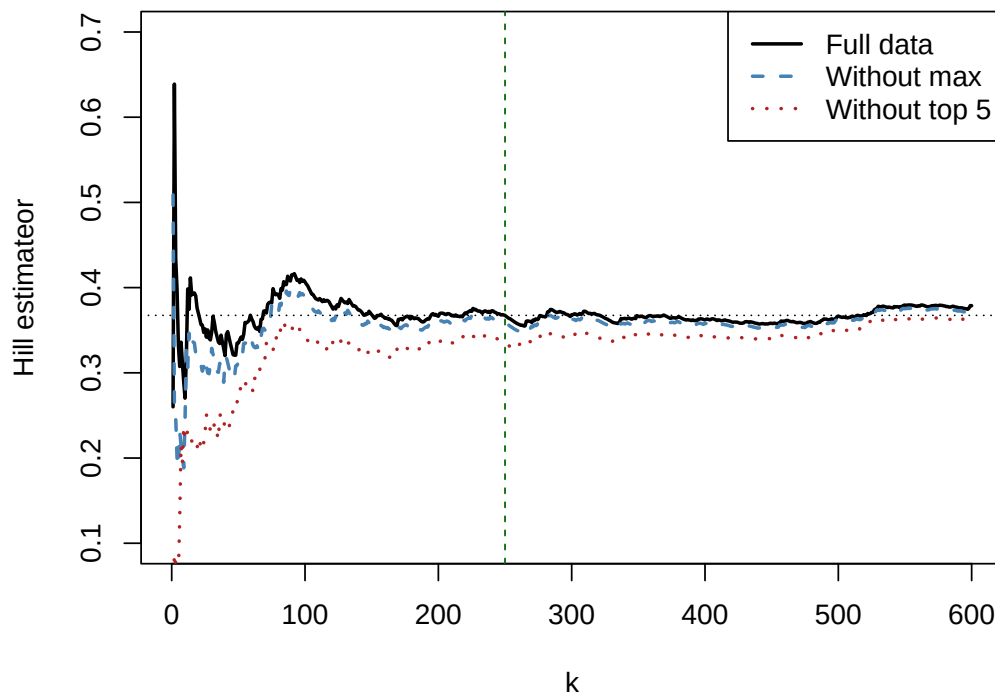


Figure 11: Hill estimator of the tree datasets

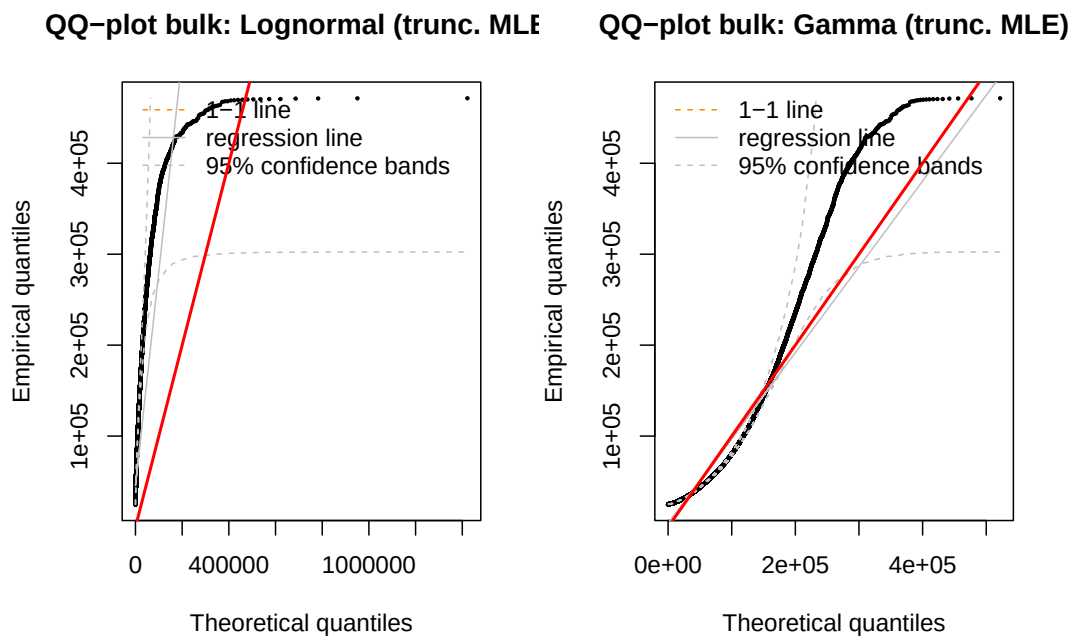


Figure 12: QQ-plots: Lognormal vs Gamma (truncated MLE)

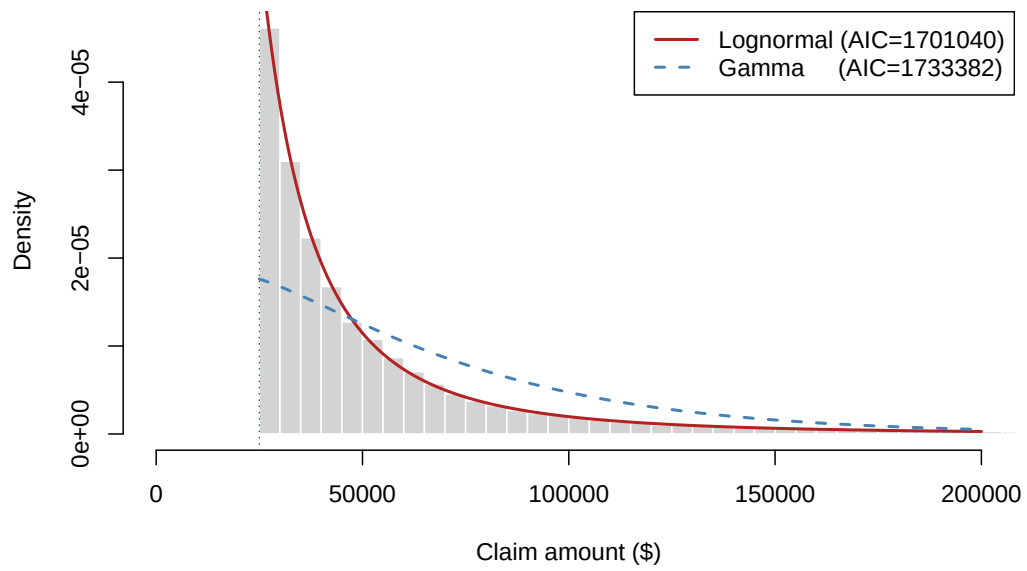


Figure 13: Bulk density: Lognormal vs Gamma (truncated MLE)